Cardboard blaster for AR laser tag

Template for iPhone 17 Pro, iPhone 14 Pro, Redmi A5 and phones up to 175 x 82 mm with the case

How to print

Print on A4 at 100% (choose "Actual size" when printing, not "Fit to page").
Check the square below with a ruler: it must measure 50 mm.
Print one copy only: cut out the paper template and trace it onto the cardboard as many times as you need. One set is enough for 4 blasters.

Materials for one blaster

Corrugated box cardboard (3-4 mm), about 2 medium boxes.
PVA glue or hot glue (an adult uses the hot glue).
3 skewers, 1 toothpick, hair ties and 2 large rubber bands, copper tape, sponge, a small piece of sponge (kitchen sponge, soft side), sticky tape, cord for the wrist strap. Craft knife and cutting mat: an adult uses them.

Parts

Side: 3 layers (pages 2 and 3, joined): the middle layer also has the lever slot.
Lever, cheeks, end caps, low lips, brackets (page 4). Phone box (page 5).
Assembly diagram and steps (page 6).

Key

Black line: cut. Dashed orange: cut only where shown, or fold.
Circle with cross: hole (pivot, toothpick, wrist strap).

Safety

The phone always keeps its case on. Strap on the wrist, never around the neck.
Walk, don't run, on a rug or in a clear room.
Before playing, gently shake the blaster upside down over the sofa: the phone must not move.

50 mm

if it is not 50 mm,
print again at 100%

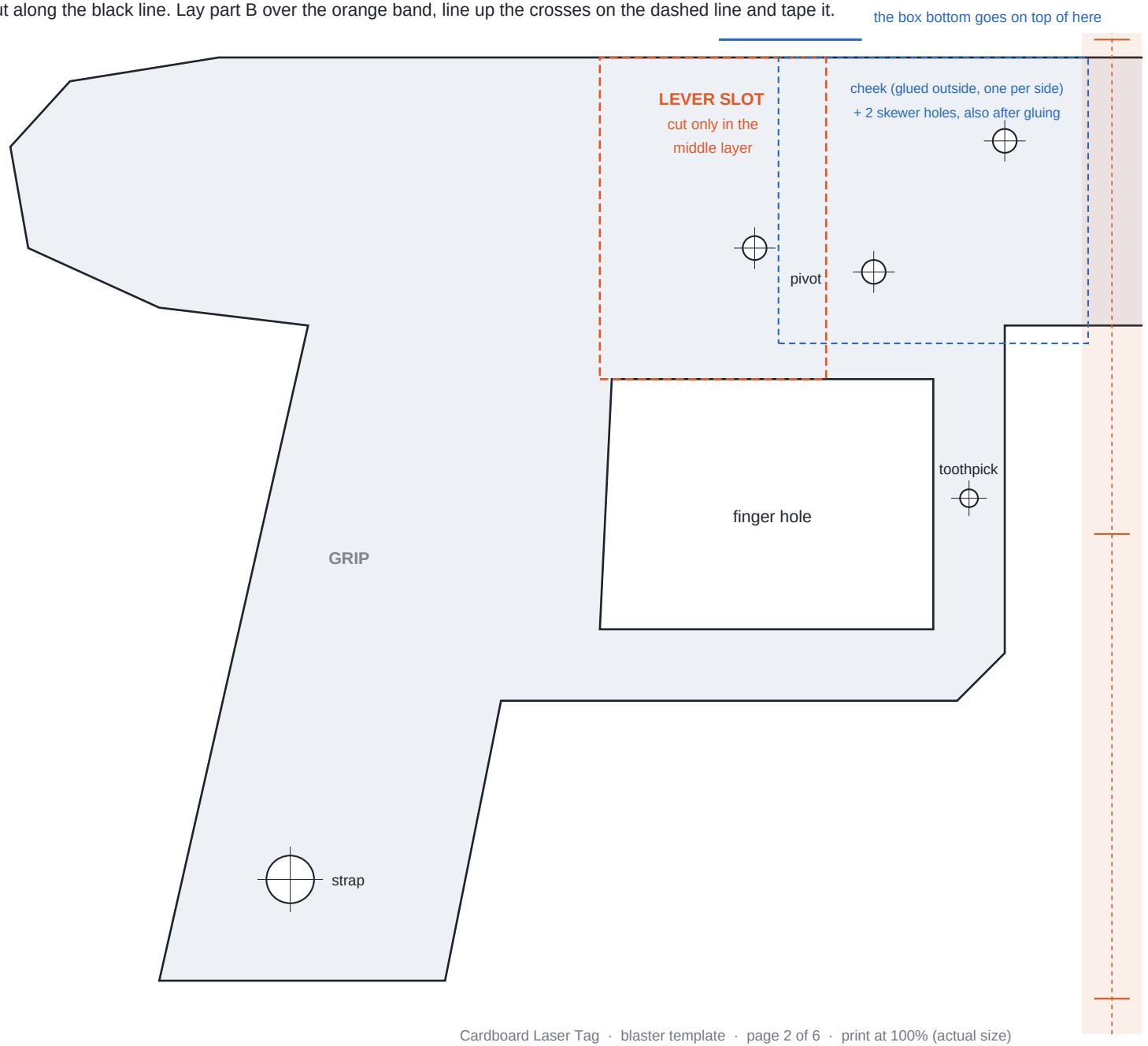
Scale check

Measure the side of the square before cutting.

A 3% error is enough for the phone not to fit.

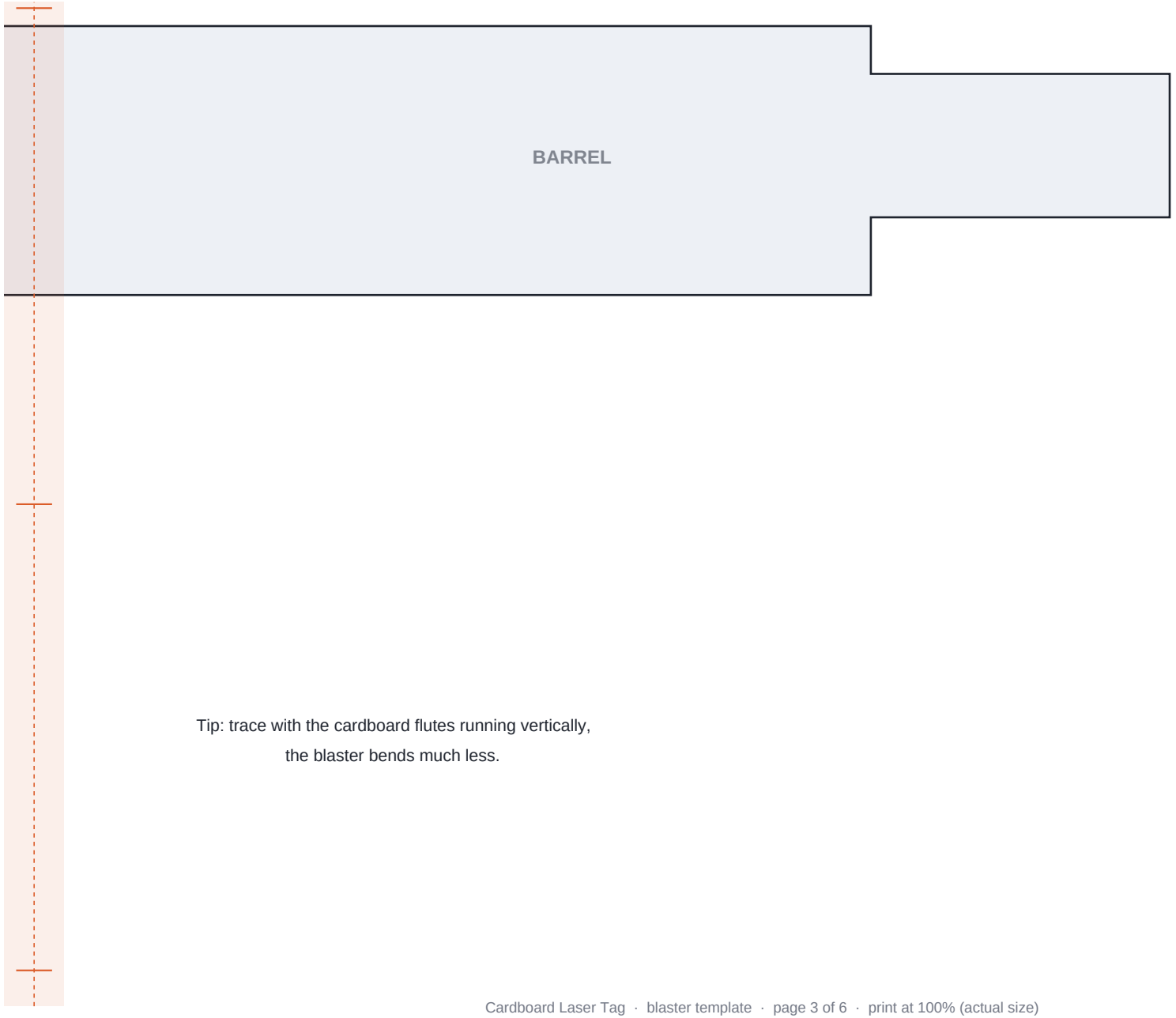
Side, part A of 2

Cut along the black line. Lay part B over the orange band, line up the crosses on the dashed line and tape it.



Side, part B of 2

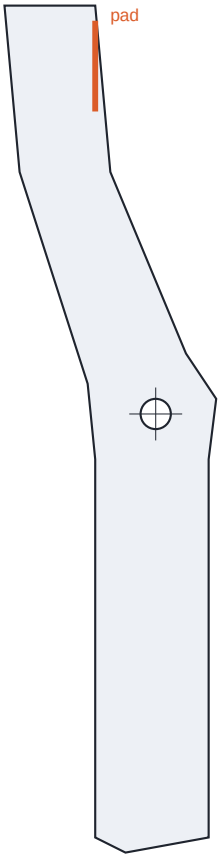
Cut along the black line and join to part A. Use the whole side as a stencil: trace it 3 times on thick cardboard.



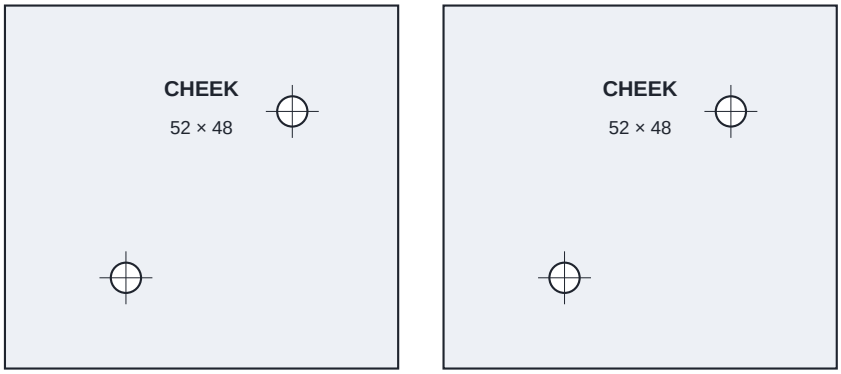
Tip: trace with the cardboard flutes running vertically, the blaster bends much less.

Small parts

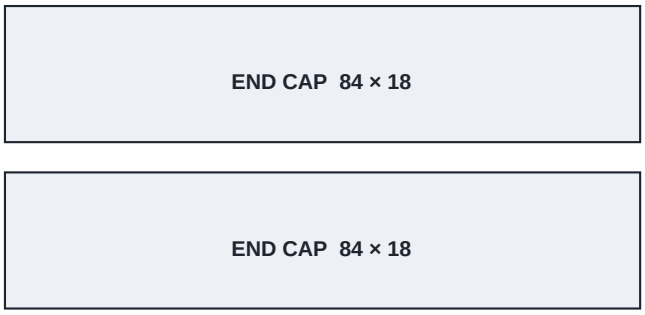
All in thick cardboard.



LEVER ×1
flutes vertical



×2, one per side. The two holes are for the skewers:
drill them together with the side, with the cheek in place.

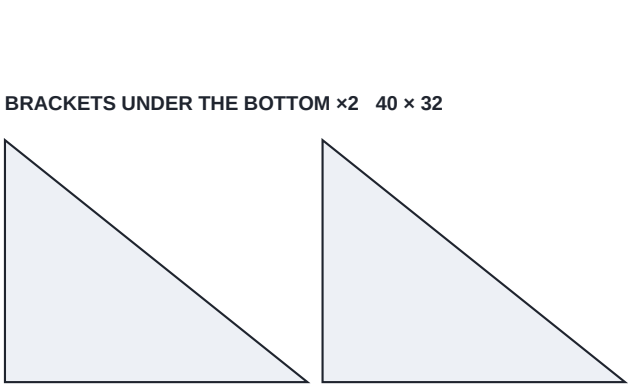


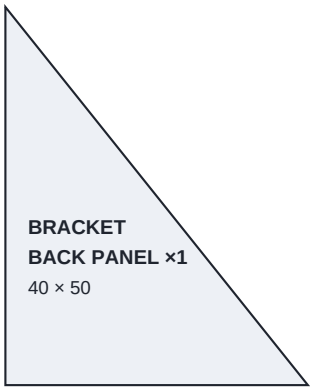
×2, they close the ends of the box (long side standing)

×2, in front of the screen at the bottom, on the sides

Sponge: two strips of soft
sponge about 1 cm thick:
55 × 50 on the back panel,
150 × 10 under the lid.

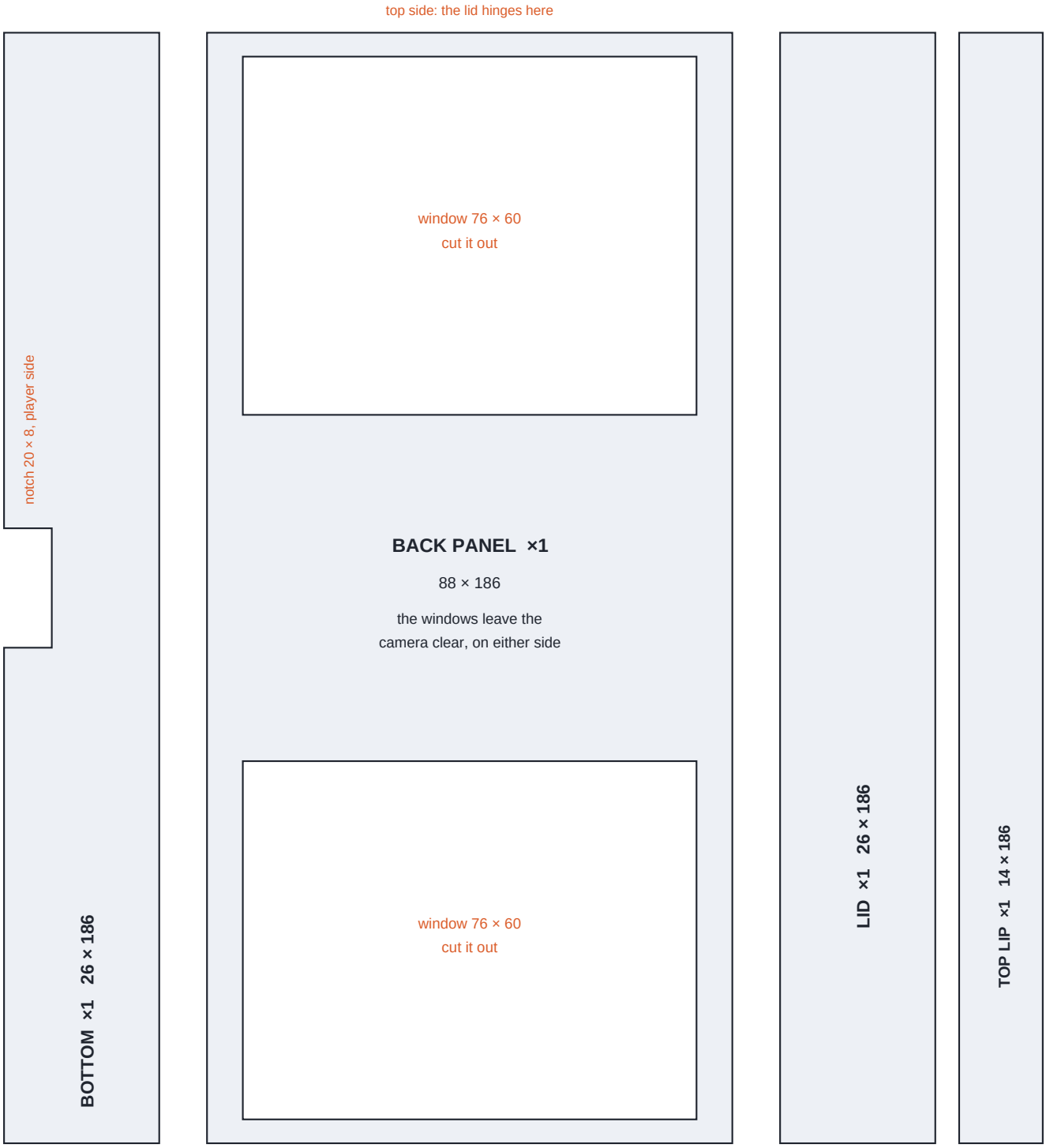
BRACKETS UNDER THE BOTTOM ×2 40 × 32





Phone box

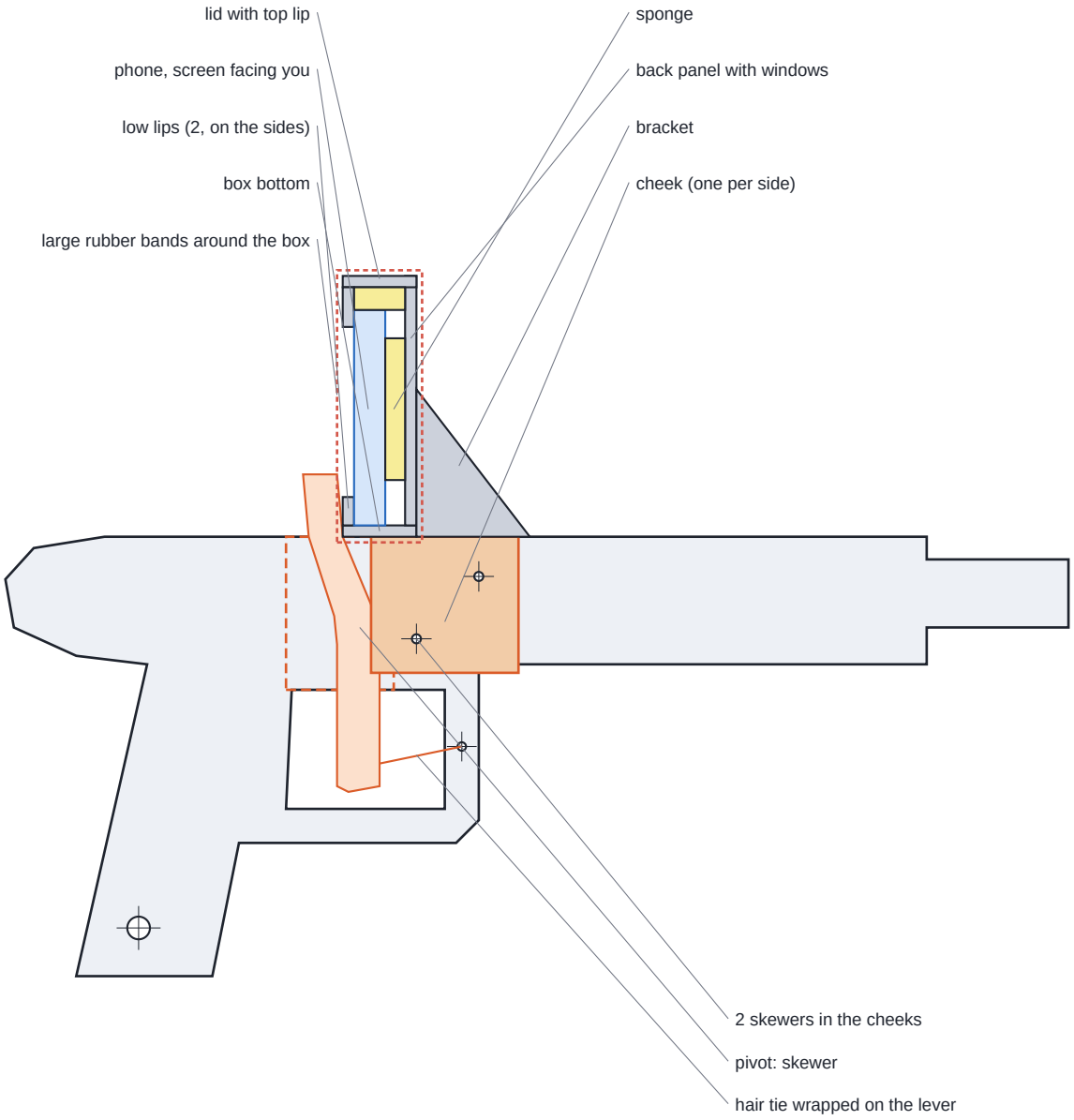
Long pieces, all in thick cardboard, flutes horizontal (along the piece).



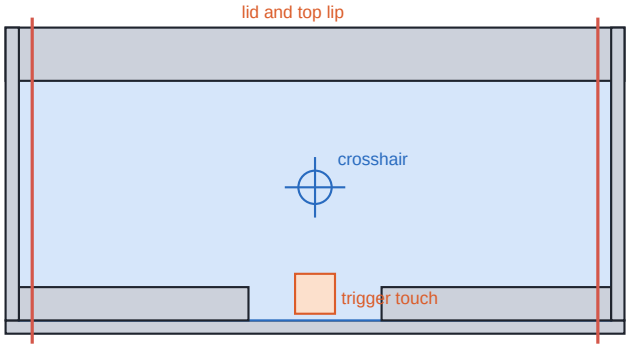
Inside of the box: 178 × 84 × 18 mm. It fits the biggest phone (Redmi A5 with case); for smaller phones the sponge fills the gap. The ends of the box stay open behind, in the windows: the camera can always see, whatever the phone.

Assembly diagram

Side view (scale about 1:2.5)



Back view: what you see



Steps

1. Side: 3 layers, lever slot only in the middle one.
   Holes: pivot, toothpick and strap through all layers.
2. Lever: copper on the front, from trigger to tip, pad on the tip.
3. Glue the 3 layers with the lever in its slot, skewer as pivot,
   hair tie wrapped on the lever and around the toothpick.
4. Cheeks on both sides of the side piece, PVA glue, flat.
   Drill and push the 2 skewers through cheeks and side.
5. Bottom on top of cheeks and side, brackets under the bottom.
6. Back panel, end caps and low lips on the bottom, hot glue.
   Bracket behind the back panel. Sponge on the back panel.
7. Lid: tape hinge on the top edge of the back panel,
   top lip glued on the front, sponge underneath.
8. Phone in from the top, lid down, 2 large rubber bands
   around the ends of the box.
9. Test: the trigger must touch the screen at the bottom centre.